THE CHANGE-MAKERS

TATA MOTORS
At the start of the calendar year, Tata was selling about 3,500 EVs a month; by the end of July, the number grew to 6,500. It is adding more electric models

MAHINDRA
This Independence Day, Mahindra showcased the electric variant of their premium off-roader, Thar. It is looking at bringing in new variants of its SUVs

TESLA
Although Tesla still doesn't have much of a stake in the Indian market, being at the forefront of EV innovation, it raises the bar for domestic players

10 MILLION
Expected units of annual sale of EVs in India by 2030, per Economic Survey, 2022-23

PRAVAIG DYNAMICS
Founded in 2011, this deep tech mobility firm is making EVs, batteries and AI solutions for autonomous vehicles



"The future is in controlling data. Unlike conventional automobiles, connected EVs generate data and are dependent for their smooth functioning and expansion on that data. Our own data is being used against us at an individual level for digital rent seeking.... We need to take back control now"

– SIDHARTH BAGRI
Founder & CEO, Pravaig Dynamics

EV manufacturing. While CNG, hybrid fuels and experiments with hydrogen are viable alternatives in the transport sector, EVs are deemed to have the greatest potential.

WHY IT IS A GAME CHANGER

A widespread presence of EVs on Indian roads would give a fillip to the overall acceptability of renewable energy as a source for baseload (24-hour) power. The Economic Survey envisaged "carefully-crafted, multi-dimensional mineral policy" that would reduce India's dependence on foreign sources for rare earth elements essential for EVs, along with production-linked incentives for setting up advanced lithium-ion cell and battery manufacturing capacities. India aims for a gradual increase of production to 50 GWh (Gigawatt-Hour) battery capacity through the scheme—titled National Programme on Advanced Chemistry Cell (ACC) Battery Storage—between FY23 and FY27. This is expected to act as a spur to establish a flourishing battery/ cell manufacturing industry. An electric vehicle uses one or more electric motors for propulsion and can be powered by a collector system (an overhead electric line, for instance), or autonomously by a battery.

Unlike the ones used with fossil fuel, batteries that power EVs are heavier and have low energy output—a potential problem with trucks, trains or buses that need significant energy to move a load. If there is an inexorable shift towards EVs, this will result in a race to cut down on battery size to make them more efficient and cost-effective.

WHAT INDIA HAS TO MASTER

Along with cutting down on battery cost and local manufacture of EV parts, large-scale adoption of EVs requires a wide network of charging stations. What we have right now is insufficient. India's automobile sector contributes to roughly half of its manufacturing. Conventional vehicles have around 3,000 moving parts; EVs have 70. India, with its robust auxiliary manufacturing ecosystem, should make that transformation easily. ■